2-Mic Sound Locator: Wiring Guide

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Two Adafruit PDM mics on one DAT wire, a DE2-115, and a VGA screen that shows each mic's level and frequency, the direction of the sound as an image, and the nearest mic.

Parts and project files

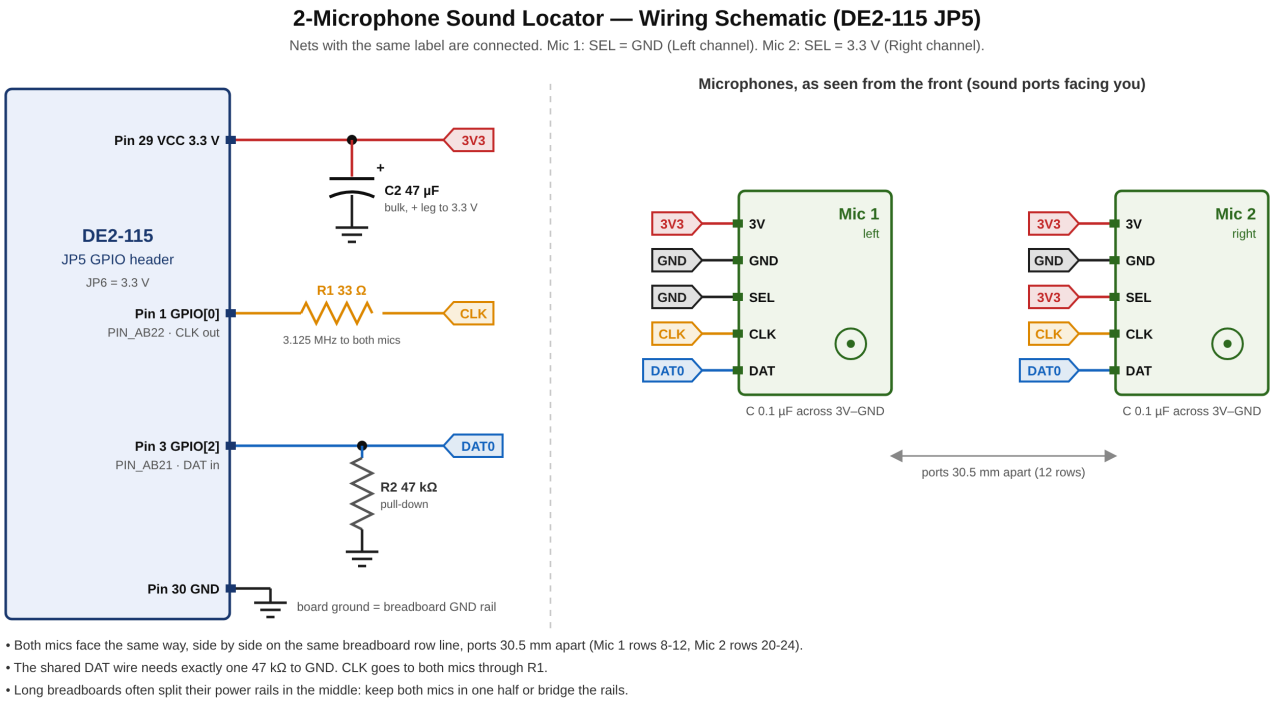
The circuit is your working single-mic circuit plus one more mic on the same wires; no new FPGA pins are used.

Parts: 2 Adafruit PDM MEMS mics (MP34DT01-M), 1 x 33 ohm, 1 x 47 kohm, 2 x 0.1 uF, 1 x 47 uF, breadboard, 5 short female-to-male jumpers, VGA monitor and cable, a phone with a tone-generator app.

File	What it is
de2_115_two_mic_top.v	Top level: mic capture, two channels, direction sweep, history image, LEDs/HEX
vga_2mic_display.v	VGA screen: sensor panel, source panel, direction image, mic icons
pdm_channel.v	One mic channel (same as v3, verified chain)
mic_ring_buffer_1k.v	1024-sample history per mic
freq_counter_hz.v	Frequency counter (same as the single-mic test)
pin_assignments_2mic.tcl	Quartus pins: mic, VGA, switches, LEDs, HEX
two_mic.sdc	Timing constraints (one 50 MHz clock)
tb_two_mic.v, run_sim_2mic.do	ModelSim testbench (saves VGA screenshots) and script

Wiring

Mic 1 sits on the left with SEL to GND, Mic 2 on the right with SEL to 3.3 V, and both drive the same DAT wire on JP5 pin 3.



Signal	Mic 1 (left)	Mic 2 (right)	DE2-115 JP5
3V	+ rail	+ rail	Pin 29 (3.3 V)
GND	− rail	− rail	Pin 30 (GND)
SEL	− rail (GND)	+ rail (3.3 V)	—
CLK	shared CLK row	shared CLK row	Pin 1 through the 33 ohm resistor
DAT	shared DAT row	shared DAT row	Pin 3, with 47 kohm from DAT to GND

1. Power the DE2-115 off. Leave the working Mic 1 wiring as it is.
2. Insert Mic 2 in the same orientation as Mic 1 (pin labels readable the same way), with its header 12 rows to the right of Mic 1's. That puts the two sound ports 30.5 mm apart, which the code assumes.
3. Put a 0.1 uF capacitor across Mic 2's 3V and GND columns.
4. Wire Mic 2's 3V to the + rail and GND to the − rail.
5. Wire Mic 2's SEL to the **+ rail (3.3 V)**. This is what makes Mic 2 talk in the other half of the clock cycle.
6. Jumper Mic 2's CLK column to Mic 1's CLK column (the column the 33 ohm resistor feeds). Do not add a second 33 ohm.

7. Jumper Mic 2's DAT column to Mic 1's DAT column. Do not add a second 47 kohm; one per DAT wire.
8. Check both mics are in the same half of the breadboard, or bridge the rails if your board splits them in the middle.
9. Connect the VGA monitor to the DE2-115 VGA port.

Checks before powering on

With the board off and the meter on continuity (·)))), every row below should match before you power up.

Probe between	Expect
Mic 1 SEL ↔ Mic 1 GND	Beep
Mic 2 SEL ↔ Mic 2 3V	Beep
Mic 1 CLK ↔ Mic 2 CLK	Beep
Mic 1 DAT ↔ Mic 2 DAT	Beep
Mic DAT ↔ JP5 pin 3 jumper end	Beep
Mic 3V ↔ Mic GND (either mic)	No steady beep (a short chirp is the 47 uF charging)
Mic DAT ↔ Mic CLK	No beep

Then power on and set the meter to V⎓: both mics' 3V pins read 3.2–3.3 V against GND, Mic 1 SEL reads 0 V and Mic 2 SEL reads 3.3 V.

Optional board check before the VGA build: load the single-mic test again. SW0 down should show Mic 1 and SW0 up should show Mic 2, both with density about 80 on HEX7–6.

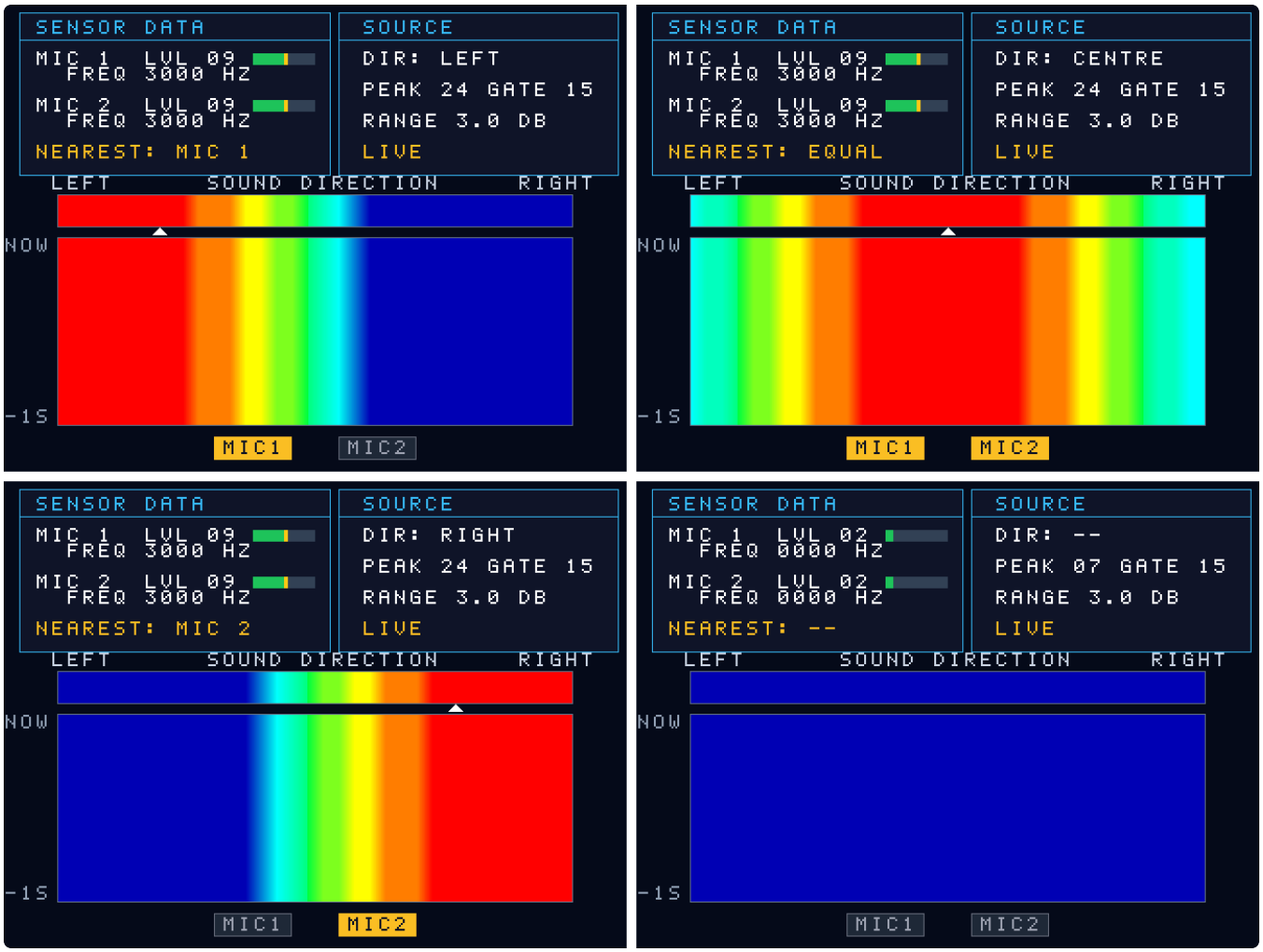
Build and program

1. New Project Wizard: name `two_mic`, top-level entity `de2_115_two_mic_top`, device EP4CE115F29C7.
2. Add the five .v files (top, vga_2mic_display, pdm_channel, mic_ring_buffer_1k, freq_counter_hz) and `two_mic.sdc`.
3. Tools > Tcl Scripts > `pin_assignments_2mic.tcl` > Run.

4. Compile. Check that TimeQuest shows no failing paths at 50 MHz and the Fitter uses about 12 M9K RAM blocks.
5. Program the `.sof` over JTAG (re-program after every power cycle).

What the screen shows

The sensor data that was on the 7-segment displays now sits in the top-left panel, and the sound appears as a colour image of direction over time with the nearest mic highlighted.



The preview above is rendered from the same layout and font tables as the Verilog: tone on Mic 1's side (top-left), straight ahead (top-right), on Mic 2's side (bottom-left), and silence (bottom-right).

Screen area	Shows
SENSOR DATA (top-left)	Each mic's level (LVL, 1 step = 6 dB) with a bar, its frequency in Hz, and NEAREST: MIC 1 / MIC 2 / EQUAL / --
SOURCE (top-right)	Direction (LEFT, CENTRE, RIGHT), loudness of the loudest direction against the noise gate (PEAK, GATE), colour range, LIVE or HOLD, MIRROR
Live bar	Intensity across 33 directions, far left to far right; white marker under the loudest direction
History image	The same bar every ~22 ms, newest at the top, about 1 s shown; a moving source draws a track
MIC1 / MIC2 icons	Nearest mic filled amber; both amber when EQUAL

Control / indicator	Function
KEY0	Reset
SW1, SW0	Colour range below the peak: down/down 3 dB (default), down/up 6 dB, up/down 12 dB, up/up 1.5 dB
SW5–SW2	Raise the noise gate (GATE = 15 + switch value)
SW16	Mirror left-right (use if the array faces away from you)
SW17	Hold the history image
HEX7–6, HEX5–4	Mic 1 and Mic 2 level (hex)
HEX3–0	Frequency of the nearest mic, Hz
LEDG0, LEDG1	Mic 1, Mic 2 data healthy
LEDG8	Sound above the gate
LEDR7–0, LEDR15–8	Mic 1 and Mic 2 level bars

First test

A 3 kHz phone tone held to one side should turn that side of the image red and highlight the mic on that side.

1. Check LEDG0 and LEDG1 are on. In silence, the panel should read LVL about 02–04, FREQ 0000 and NEAREST: --.
2. Stand the breadboard up so the sound ports face you. Mic 1 should be on your left.
3. Play a steady 3000 Hz tone 20–40 cm away, level with the mics. Both FREQ lines should read 3000 HZ.
4. Move the phone to the far left: the left of the image turns red and MIC1 lights. Far right: MIC2. Straight ahead: centre red and NEAREST: EQUAL.
5. Sweep the phone slowly left to right and back; the history image draws the path.

Use tones from 2 to 4 kHz. Below 2 kHz two mics 3 cm apart cannot separate directions, and above about 5 kHz the direction becomes ambiguous.

Symptom	Likely cause	Fix
One LVL stays at 00–02 and its LEDG is off	That mic is not powered, or its SEL is wrong	Recheck the pre-power table for that mic
Both mics' readings jump around, LEDG0/LEDG1 flicker	Both SEL pins tied the same way, so the mics fight on DAT	Mic 1 SEL to GND, Mic 2 SEL to 3.3 V
Image stays blue, NEAREST: --	Sound below the noise gate	Move closer or lower SW5–SW2; PEAK must reach GATE
Whole bar red, no clear side	Tone below 2 kHz, or range too wide	3 kHz tone, SW1/SW0 down
Left and right swapped	Array facing away from you	Turn it round, or flip SW16
No picture	VGA cable or monitor input	Same VGA pins as your earlier designs

How it works and verification

The mic interface and filters are exactly the chain that passed on your board in the single-mic test; the new parts are the direction sweep and the VGA screen.

1. **Capture.** The FPGA makes the 3.125 MHz clock. Mic 1 (SEL = GND) drives DAT while CLK is high and is read late in that half; Mic 2 (SEL = 3.3 V) drives it while CLK is low and is read late in that half.
2. **Channels.** Each mic: CIC decimator to 195.3 kHz, DC blocker, low-pass (~3.9 kHz). Level and frequency come from this signal. A ~1 kHz high-pass copy feeds the direction sweep.
3. **Direction sweep.** For 33 directions, Mic 1's signal is delayed by −16 to +16 samples relative to Mic 2 and the two are added; the energy over 512 samples (2.6 ms) is largest when the delay matches the sound's arrival difference. One sweep takes 0.35 ms.
4. **Image.** Energy becomes a log value, is smoothed over ~22 ms, and is coloured relative to the loudest direction. One row of the history image is stored every 64 sweeps.
5. **Nearest mic.** The loudest direction is taken as the middle of the near-peak run of columns; left of centre means the sound reached Mic 1 first (MIC 1), right of centre MIC 2, within one column of the centre EQUAL.

A bit-accurate Python model of this chain, driven by two sigma-delta mic models 30.5 mm apart, gave these results:

3 kHz tone at	Peak column (0–32)	NEAREST	Image active
50° to Mic 1's side	6	MIC 1	yes
30° to Mic 1's side (3.5 kHz)	8	MIC 1	yes
Straight ahead	16	EQUAL	yes
50° to Mic 2's side	25	MIC 2	yes
Silence	—	--	no

The model caught one problem that is fixed in the code: with a flat beam several columns tie for the maximum, and taking the first one pushed a centred source to column 11 (MIC 1). Using the middle of the tied run fixed it.

The Verilog itself has not been simulated here (no simulator could be installed in this workspace). Run `do run_sim_2mic.do` in ModelSim first: it prints the peak column and nearest mic, and saves `frame_left.ppm` and `frame_right.ppm`, screenshots of the VGA output you can open in any image viewer.